

Introduction

This report aims to outline a suitable proposal for an IT system infrastructure, giving insight into the overall aims of the solution and how they will be achieved. Then, the report delves further into the specific requirements and proposed technologies for the Network, followed by the Security of the infrastructure.

System Architecture

DirectSales UK's new regional branch in Plymouth requires the design and implementation of a robust and scalable network infrastructure. The core infrastructure proposal should detail an efficient system architecture for DirectSales UK with a focus on seamlessly integrating affordable, robust and scalable networking technologies and security systems. This will include routers, switches, firewalls, intrusion prevention systems, as well as other network strategies and technologies such as IP addressing, subnetting, VLANs, LAN switching, and routing. The core infrastructure designs will facilitate any future expansions to the Plymouth branch while maintaining performance, reliability and security. The infrastructure will support both wired and wireless connectivity, meeting and building upon the initial requirements of the solution: Staff PCs, staff Wi-Fi, and guest network (Admin – 15 hosts, sales team – 35 hosts, guest network Wi-Fi – 55 hosts, employee Wi-Fi - 35 hosts). Company's web and e-commerce systems (20 hosts) The company also plans to increase its sales team further in two years' time.

Listed are the Key Performance Indicators for the proposal:

- Network infrastructure should accommodate the growing needs of the business by taking a scalable approach to network design.
- Continued availability and minimal downtime to support business operations.
- System and network security implementation is sensibly priced and provides sufficient protection.
- Should deliver high speed wireless and wired connections
- Simplistic, central monitoring and management of the system allowing for Network engineers to quickly identify and fix issues

Network Architecture

This section of the proposal addresses the key components needed to create a scalable network infrastructure such as IP addressing, subnetting, switching and VLANs.

IP addressing allows connectivity of employee PC's, staff Wi-Fi and wireless access points for the guest network Wi-Fi. When establishing a communication link, the hosts interface receives an IP address from the router. One router interface sends this IP address to the host, the other receives an IP datagram containing addressing and control fields in its header and the data that is being sent over the network in its payload. The proposed solution should allocate IPV4 and IPV6 addresses efficiently for devices on the network. IPV4 uses 32-bit addresses, has a low overhead, does not always guarantee data delivery and is unable to manage or recover data losses. IPV6 uses 128-bit addresses and is being introduced to replace IPV4 when it runs out of available addresses. Currently, IPV4 is the standard for the internet and exists alongside IPV6. Header size is smaller in IPV6 addresses and provides better security than its predecessor. Each have their uses on a modern network and should be allocated accordingly, IPV6 handling connections that prioritise security and IPV4 for the menial.

Subnetting is the process of splitting an address block to create multiple smaller networks, known as subnetworks or subnets, each with their own unique network address. Subnetting saves resources by efficiently allocating IP addresses and optimizing network performance. The references section of this report contains the designs for subnet routing tables in line with the network requirements. These designs show the subnet requirements to successfully implement an IP routing system for DirectSales UK. The current requirements for the number of employee Wi-Fi and sales team hosts already exceeds the number of available hosts for a 27-bit subnet. A 26-bit subnet is used instead. Admin designs also have unused hosts as 28-bit subnets only allow for 14 hosts and DirectSales UK require 15 admin hosts. The same is true with employee Wi-Fi and server hosts. The benefit of using subnet with unused hosts is that the increase in hosts needed when the sales team is increased is already prepared for, generating a scalable network approach. However, these unused hosts are wasted space on the network which can create more traffic, wasting resources and can increase the risk of a cyber-attack.

LAN switching generates efficient data transmission within the local area network (LAN). Ethernet technology is employed for its high-speeds and consistent stability, providing employees with reliable connectivity. Switches are strategically deployed to create a collision free environment by segmenting the network and controlling the flow of data through dynamic routing. Dynamic routing is the most suitable routing option as it uses real time data and adapts to changes in the network topology, finding the optimal routing paths based on current network traffic. Similarly, virtual local area networks (VLANs) segment the network, improving security and performance. These logically group users regardless of physical location by forwarding through a routing device, allowing users in different departments to interact.

Security Architecture

Cyber risk management is an essential stage in understanding the requirements of DirectSales UK's system architecture. Identifying and understanding the threats, risks, and vulnerabilities to IT systems within the company will ensure a robust network. Security concerns can arise internally and externally in any business with an IT infrastructure. Internal security concerns can include vandalism and insider threats. While external security concerns can include Dedicated Denial of Service (DDoS) attacks, Social Engineering attempts, and malicious software such as Ransomware, Malware and Trojans. These can often go unnoticed. Therefore, the prevention of external threats, risks and vulnerabilities is imperative. To mitigate the threats to DirectSales UK IT infrastructure, an effective security architecture should be implemented. Assessing security concerns allows for the prioritization of technologies appropriate for the solution.

As a baseline, transactions like those in a direct-to-consumer e-commerce system should use encryption software to protect data transfer and provide a service that does not jeopardise sensitive client or company details. Laws like the Data Protection Act (2018) require that DirectSales UK handle user data appropriately.

Employees should receive training regarding how to identify and deal with social engineering attempts as these types of attacks are commonly deployed against businesses. This training should be mandatory upon induction and reviewed by each employee every few months to ensure individuals remember their responsibilities when interacting with the internet.

One technology capable of providing network security is firewalls. By filtering traffic between different networks, acting as a barrier, firewalls provide protection from harmful services, controlling access to site systems. Types of firewalls achieve this in unique ways.

Packet filtering gateways are a common type of firewall that operate at IP packet level within a network. This stateless firewall treats each packet individually controlling incoming and outgoing packets. Effectively, this acts as an access control system for the internal network. Stopping unauthorised machines from accessing the internal network and limiting the internet services that users can access. Configuring the access privileges could enhance employee productivity and prevent users from accessing dangerous content or infecting the network. Packet Filtering can be implemented quickly across the whole network at once making it easy to implement. It also does not require any additional hardware or software as it is part of the standard router configuration making it an affordable option. Despite these qualities, packet filtering may not be the best option for DirectSales network as it can be highly susceptible to the Denial-of-Service Attacks.

Stateful inspection firewalls share qualities with packet filtering gateways. Unlike packet filtering gateways every packet compared to a pre-defined rule set. Once the first packet has passed this ruleset, all associated packets are passed through without needing to be checked. Comprehensive packet analysing grants more advanced security configuration options and better protection against attackers. Stateful Inspection Firewalls are a more appropriate for the security requirements despite more intensive CPU usage and increased costing.

Application proxy firewalls are one of the best options for protecting a network. Providing a proxy for accessing external systems, users do not connect to or interact directly with the internet. Instead, the proxy represents the user, granting anonymity from outside services, filtering external data as it enters the network. Application proxies can act in addition to packet filtering giving greater control of the connections sent to the application gateway. Using an application gateway, all services that do not use a proxy are blocked. All authentications occur at the gateway, so if a packet is not authorised, the connection will never make it past and onto the internal systems. Software and hardware for application gateways only need to be implemented at the gateway. Therefore, maintainability of the system would be easier and more affordable. A decentralised application proxy allows for a more scalable approach to networking. This will ensure that the network can accommodate any increase in users, for instance, the sales team increasing. This can be achieved by adding more proxy nodes to handle the increase in traffic. Further benefits of a decentralised implementation include performance and fault tolerance. Based on these factors, an application proxy would be an appropriate implementation choice.

Firewalls do not always guarantee system security. Vulnerabilities can arise in the event of an error, or a mistake when designing or configuring the firewall. Attackers may exploit these vulnerabilities. A report by Kamara et al. (2003) summarises the main causes of firewall vulnerabilities: validation error, serialisation error, boundary checking error, domain error, incorrect designs, illegitimate code execution, illegitimate access to target resources and denial-of-service. The report also finds 'common firewall vulnerabilities and misconfigurations include: (1) ICMP allowed, e.g., the firewall can be *ping*-ed; (2) *Denial* rather than *drop* of traffic to ports the firewall is blocking. ("Analysis of vulnerabilities in Internet firewalls - ScienceDirect") This provides the attacker with additional information, or improves the speed of the attacker's port scan; (3) Misconfiguration that allows a TCP ping of internal hosts with Internet-routable IP addresses...; (4) Trust of certain IP addresses; (5) Availability of extra/unnecessary services on the firewall; (6) Unnecessarily open TCP and UDP ports; and (7) Firewalls that resolve to 'firewall.client.com' 'gateway.client.com' or something equally conspicuous.' The report goes on to further analyse the causes and effects of these vulnerabilities.

If a network firewall fails, and an intruder slips past the unnoticed, an Intrusion Detection system (IDS) or Intrusion Prevention system (IPS) would be the next logical line of defence. IDS' and IPSs exist on the network and analyse the traffic on the network, alerting in the event of abnormal traffic. Intrusion detection simply finds these intruders, whereas Intrusion Prevention builds upon this and will find and attempt to prevent harmful traffic by denying the connection.

Security measures must be in place for external networks attempting to access DirectSales UK's internal network. 'In networking, Demilitarized Zone is a physical or virtual network or sub-network which separates internal network usually LAN & WAN from untrusted external network, i.e. Internet. By doing so, it provides an additional layer of security to any network as hackers are unable to access internal services directly using internet' (Shrimali, 2017). Web and E-commerce systems will be an interaction point for external networks. Using a DMZ will reinforce the security of the network, cutting them off from the internal network should these applications become compromised. The web and ecommerce systems should not be entirely dependent on network defences and should contain built in protection from web attacks such as SQL injection.

Threats to infrastructure can also come from within the company itself. A disgruntled or former employee may intentionally inflict harm to the digital systems. Risks can be reduced by employing the least privilege principle, where employees are only given the access rights necessary to effectively do their jobs. The access rights can easily be managed using Role Based Access Control (RBAC). It is important that users maintain the rights corresponding to their roles and users are removed from roles if they leave the company. This applies to the physical security of the network. Preventing accidental damage or attacks from malicious employees, the servers and any other technology key to the operation and security of the network should be kept in a dedicated space, kept locked to unauthorised employees. Access should be granted to only the essential staff, protected through biometrics. Biometrics is favourable over a token-based authentication access card or key. A physical key could be misplaced, stolen, or reproduced. Biometrics is more expensive to implement, an affordable alternative is pin based access. Employee's physical and digital access rights should be frequently assessed to uphold security standards. By keeping servers behind locked doors, the risk of accidental damage to the network is mitigated.

Considering business expenses and requirements of the system, the following technologies are justified for use in DirectSales UK's security architecture: Staff training equips employees to identify and avoid security threats, while application proxy firewall, demilitarised zone, and Intrusion Prevention System provide robust technical defences against cyberattacks. Implementing these technologies will maximise the security of the network and user safety while ensuring affordable deployment and maintainability.

Conclusion

In conclusion, the IT system infrastructure proposed for DirectSales UK will grant a comprehensive approach to network and security that fulfils the company's requirements while accommodating future development and expansion. The proposal effectively provides a safeguard from threats to the system while limiting the costs.

Through IPV4 and IPV6 addressing, efficient subnetting and strategic LAN switching, the network infrastructure will support the company's growth while maintaining optimal performance. The use of VLANS further enhances the security and network management of the system.

The multi-layer approach to security architecture, including encryption software, firewalls, and intrusion detection/prevention systems demonstrates a robust defense against cyber threats. The selected firewall type works well for DirectSales UK's requirements. Thorough consideration of physical and cyber security requirements will prepare for internal and external threats.

Overall, the proposed IT system infrastructure for DirectSales UK not only meets the company's immediate requirements but also establishes a foundation for future growth. By implementing the recommended technologies and security measures, DirectSales UK can safeguard its operations and data while ensuring affordability and scalability.

References

Kamara, S., Fahmy, S., Schultz, E., Kerschbaum, F. and Frantzen, M. (2003). Analysis of vulnerabilities in Internet firewalls. *Computers & Security*, [online] 22(3), pp.214–232. doi:[https://doi.org/10.1016/S01674048\(03\)003109](https://doi.org/10.1016/S01674048(03)003109).

Shrimali, S. (2017). DeMilitarized Zone: Network Architecture for Information Security. *International Journal of Computer Applications*, [online] 174(5), pp.16–19. doi:<https://doi.org/10.5120/ijca2017915389>.

Data Protection Act 2018.c.12. Legislation.gov.uk. [online] Available at: <https://www.legislation.gov.uk/ukpga/2018/12/contents/enacted>.

Subnet Routing Tables

Network: 17.1.7.0 / 24

Admin

Specification	Value
Number of bits in the subnet	27
New IP mask (decimal)	255.255.255.224
Number of usable subnets	8
No. of usable hosts per subnet	30
Network address	17.1.7.0
First IP Host address	17.1.7.1
Last IP Host address	17.1.7.30

Sales

Specification	Value
Number of bits in the subnet	26
New IP mask (decimal)	255.255.255.192
Number of usable subnets	4
No. of usable hosts per subnet	62
Network address	17.1.7.0
First IP Host address	17.1.7.1
Last IP Host address	17.1.7.62

Guest Wi-Fi

Specification	Value
Number of bits in the subnet	26
New IP mask (decimal)	255.255.255.192
Number of usable subnets	4
No. of usable hosts per subnet	62
Network address	17.1.7.0
First IP Host address	17.1.7.1
Last IP Host address	17.1.7.62

Employee Wi-Fi

Specification	Value
Number of bits in the subnet	26
New IP mask (decimal)	255.255.255.192
Number of usable subnets	4
No. of usable hosts per subnet	62
Network address	17.1.7.0
First IP Host address	17.1.7.1
Last IP Host address	17.1.7.62

Servers

Specification	Value
Number of bits in the subnet	27
New IP mask (decimal)	255.255.255.224
Number of usable subnets	8
No. of usable hosts per subnet	30
Network address	17.1.7.0
First IP Host address	17.1.7.1
Last IP Host address	17.1.7.30